Research Engineer / Sr. Research Engineer - Mechanical Designer/Engineer for Propulsion System

Southwest Research Institute · San Antonio, TX

Job ID (this archive)	ENERGY-019
Employer	Southwest Research Institute
Location(s)	San Antonio, TX
Career stage (archive tag)	Entry or experienced (Entry / Senior)
Location group	Texas (outside DFW)
Majors (archive tags)	Mechanical Engineering, Materials Science & Engineering
Energy area	Power generation & thermal R&D
Pay listed	Not listed
Recruiting status when checked	Status not confirmed: the posting text was found, but current hiring was not verified. Checked Oct 25, 2026.
Source page	https://resapp.swri.org/ResApp/Job_Details.aspx?JOB_CD=18-01605
Original link (from the course sheet)	https://www.ziprecruiter.com/job-redirect?match_token=CqABChZQd281Yl9lOHNWMVVpbjlNN3NnQjlBEiQwMWEwY2FiMC1mMTQzLTcwOWUtOTE0Mi1mYjMyZmM2MjNjYWMaXEFBRVdYaUpqalJxd2FBX3pFSDJ6c2ZqczBwOVFJT2M2dDB4VmtGX2VkelR6X2xxb3dSSXpqSkk1bllaTGJIX3p0d1dkSno4RnNpdW5vZUluQk84RzhjUHNaMlJ2INqtBRABGNqtBQ%3D%3D&server_redirect=true&tsid=100000600
Captured	2026-10-25
Course sheet refs	ENERGY-019

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Job description

Who We Are:

The Propulsion & Energy Machinery Section performs engineering R&D in the fields of gas turbine combustion, air-breathing propulsion, industrial heat and power, and liquid propulsion. Our technologies are powering a cleaner future and advancing state-of-the-art propulsion systems for air and space flight.

Objectives of this Role:

Support the design, analysis, and testing of systems and components used in a variety of propulsion and energy processes.

Typical application areas include: supercritical CO2 oxy-fuel combustion systems, gas turbine combustion systems, and launch vehicle components.

Apply principles of machine design, piping/pressure vessel design, process controls and instrumentation, fluid dynamics, thermodynamics, and heat transfer to develop test systems and specify components.

Many systems operate at extreme pressure and temperature conditions.

Apply knowledge of fabrication methods, assembly, and geometric dimensioning & tolerancing (GD&T) to ensure components are designed in a way that can be manufactured and assembled.

Utilize CAD software (Solidworks) to develop mechanical designs and visualize component fitments and assemblies.

Communicate ideas, results, and other information to coworkers as well as commercial and government clients.

Daily and Monthly Responsibilities:

Provide design, analytical, and experimental support to more than one project in a team environment.

Typical tasks include research, design, analysis, and simulation development. Use commercial engineering software such as ANSYS Workbench or APDL for design and analysis.

Provide mechanical and machine design support for a variety of applications that include power generation, propulsion, and combustion.

As needed, work across the Institute to draw on the experience from thousands of engineers and technicians.

Contribute and be responsible for technical development of experimental test programs, including developing/operating a new test rig.

Take responsibility for testing/rig modifications to meet new project objectives.

Direct the technical work of other engineers and technicians.

Assess the best approach to solve problems and interrogate results for reliability.

Provide troubleshooting support for tests and simulations with a firm understanding of theories and concepts using a first-principles approach.

Act as the primary technical lead for mechanical designs and guide technical direction of other engineers for projects/proposal efforts.

Be a problem solver that exercises creativity and independent judgement in determining approaches to solve a wide range of technical problems.

Requirements:

Requires a Bachelors, Masters or a PhD in Mechanical Engineering, Aerospace Engineering, Chemical Engineering, or related engineering degree. 1 yr experience required with MS degree, 0 yr experience required with PhD in relevant areas.

2-5 years: Mechanical design experience applying tools like FEA to predict component/assembly stresses and evaluate fitness for service.

2-5 years: Design, commissioning, and testing experience with high pressure and/or extreme temperature (cryogenic or combustion) process systems.

2-5 years: Experience with industry standards, such as ASME Pressure Vessel Code, ASME B31 Piping Codes, machine assembly, maintenance, and repair.

MS or PhD degrees in engineering are preferred, but not required.

A valid/clear driver's license is required.

Special Requirements:

Applicant selected will be subject to a government security investigation and must meet eligibility requirements for access to classified information. Applicant must be a U.S. citizen.

Job Locations: San Antonio, Texas

For more information about this division, visit the

Mechanical Engineering

home page.

For benefits information at our San Antonio location, click

here

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For benefits information at all other locations, click

here

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An Equal Employment Opportunity Employer: race, color, religion, sex, national origin, disability, and veteran status.

If you need assistance with completing the application, please

Contact Us

Archive notes for ECS 1100

Retrieval note: Official SwRI requisition 18-01605 matches title and San Antonio. Full description recovered; application acceptance was not independently established.

Key skills listed in the archive: thermodynamics; turbomachinery; CAD; mechanical design; propulsion; experimental R&D

Archive metadata — Accessed October 25, 2026 (America/Chicago).

Recruiting status: Unverified · PDF: Complete · Source: Official employer / ATS

Source identifier: 18-01605 (may be an aggregator ID)

Retrieval note: Official SwRI requisition 18-01605 matches title and San Antonio. Full description recovered; application acceptance was not independently established.

Identity review: Title, company and location matched (see Notes). Source type: Official employer / ATS. Source identifier: 18-01605.

Sources and retrieval details

Original workbook URL

https://www.ziprecruiter.com/job-redirect?match_token=CqABChZQd281Yl9lOHNWMVVpbjlNN3NnQjlBEiQwMWEwY2FiMC1mMTQzLTcwOWUtOTE0Mi1mYjMyZmM2MjNjYWMaXEFBRVdYaUpqalJxd2FBX3pFSDJ6c2ZqczBwOVFJT2M2dDB4VmtGX2VkelR6X2xxb3dSSXpqSkk1bllaTGJIX3p0d1dkSno4RnNpdW5vZUluQk84RzhjUHNaMlJ2INqtBRABGNqtBQ%3D%3D&server_redirect=true&tsid=100000600

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